Appendix 1 solutions

1

The reduced odd denominator gives $(-8)^{2/3} = ((-8)^{1/3})^2 = (-2)^2 = 4$. Also $16^{1/4} = 2$, so $16^{-3/4} = 1/8$. No real r has $r^2 = -16$, so $(-16)^{1/2}$ is undefined over reals. Replacing it by a different nested-power expression cannot repair the original domain.

2

The law and $E(0) = 1$ give $E(x)E(-x) = E(0) = 1$. Neither factor can be zero, and dividing gives $E(-x) = 1/E(x)$. A hypothetical zero factor would make the product zero, contradicting one.

3

By the positive-base definition and change of base,

$$a^{\log_b c} = \exp((\log_b c)\ln a) = \exp\left(\frac{\ln c \ln a}{\ln b}\right).$$

Interchanging a, c gives the identical expression, proving equality.

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4

Since $48 = 2^4 \cdot 3$, $\log_2 48 = 4 + \log_2 3 \approx 5.584963$, using $\log_2 3 \approx 1.584963$. Change of base gives

$$\log_{10} 48 = \frac{\ln 48}{\ln 10} = \frac{\ln 48}{\ln 2}\frac{\ln 2}{\ln 10} = \log_{10} 2 \ \log_2 48.$$

The values differ by a fixed factor rather than being equal.

5

The domain requires $x - 1 > 0, x + 2 > 0$, hence $x > 1$. Within it, combine logarithms and use injectivity:

$$\ln((x-1)(x+2)) = \ln 4 \iff (x-1)(x+2) = 4.$$

Thus $x^2 + x - 6 = (x-2)(x+3) = 0$. Candidates are two and minus three, but only $x = 2$ meets the domain.

6

For nonnegative x, all terms after $1 + x$ in the exponential series are nonnegative, proving the inequality. Equality holds at zero. For the full real line, set $h(x) = e^x - 1 - x$. Then $h(0) = 0$ and $h' = e^x - 1$, negative before zero and positive after. Thus zero is its unique global minimum and equality occurs only there. An argument invoking the desired inequality again at negative x would be circular.

At $x = -2$, the bound gives $e^{-2} \geq -1$. Positivity of the exponential makes the inequality strict.

7

For $b > 0$, t^q is continuous, strictly increasing, and unbounded on nonnegative reals, giving a unique positive root. If $p/q = p'/q'$, the two positive candidates raised to qq' equal $b^{pq'} = b^{p'q}$. Injectivity of the positive power forces equality; reciprocals handle negative exponents.

For a counterexample, $((-8)^2)^{1/2} = 8$, whereas $(-8)^{2(1/2)} = -8$. Both are real and defined.

8

The domain is $x > 1$. A base-half logarithm decreases, so

$$\log_{1/2}(x-1) \geq 2 \iff 0 < x - 1 \leq 1/4 \iff 1 < x \leq 5/4.$$

At $x = 9/8$, the logarithm of $1/8$ is three and satisfies the inequality. At $x = 11/8$, the argument $3/8 > 1/4$, so the decreasing logarithm is below two and fails. The lower boundary $x = 1$ is outside the domain.

9

Let $c = \ln b > 0$ and choose integer $k > r$. One exponential-series term gives

$$0 \leq n^r / b^n \leq \frac{k!}{c^k n^{k-r}} \to 0.$$

For $u = \varepsilon \ln n$,

$$\frac{\ln n}{n^\varepsilon} = \frac{1}{\varepsilon}\frac{u}{e^u} \leq \frac{2}{\varepsilon u} \to 0,$$

using the quadratic term of the exponential series.

If $b = 1 + 1/n$, the derivative bound $(\ln(1+t))' = 1/(1+t) \leq 1$, together with $\ln 1 = 0$, gives $\ln(1+t) \leq t$ for positive t. Hence $b^n \leq e$, and $n^r / b^n \geq n^r / e \to \infty$. A fixed base is essential.

10

In the stated order, bit lengths are 1,4,4,5 and halving rounds are 0,3,4,4. For example $13 \to 7 \to 4 \to 2 \to 1$, while $16 \to 8 \to 4 \to 2 \to 1$. The formulas are respectively $\lfloor \log_2 n \rfloor + 1$ and $\lceil \log_2 n \rceil$.

$$L = 1000 + \ln(1 + e^{-1}) \approx 1000.313262.$$

Here $e^{-1} \in (0, 1)$, its sum with one lies in $(1, 2)$, and the logarithm in $(0, \ln 2)$. Do not compute e^{1000} first or discard the nonzero smaller term.

11

Since $\log_2 16 = 4$, $(\log_2 16)^2 = 16$, while $\log_2(16^2) = \log_2 256 = 8$. Iteration gives $16 \to 4 \to 2 \to 1$, so $\log_2^* 16 = 3$. These notations represent different operations.

References and attribution

Lin Cing, handwritten data structures notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Appendix 1.

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